

Assignment

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Subject: SEPM

Q1 Define a Project ? What is Project management
Explain various characteristics of Project.

- Project : A project is a temporary endeavour undertaken to create a unique product, service or result. It has a specific beginning and end it is usually constrained by resources, time & budget.
- Project management : Project management is a application of knowledge, skills, tools and techniques to project activities to meet the project requirement. It involves planning, executing, monitoring and closing a project requirement. It involves closing a project to achieve specific goals within the given constraints.
- Characteristics of project
 - 1) Temporary: Every project has a finite lifespan with defined start & end.
 - 2) Unique Purpose: A project is not a routine operation it is meant to achieve a specific unique goals.
 - 3) Progressive elaboration: Projects start with broad ideas and are developed in more detail and specifically as they progress.

- 4) Resources : Projects require resources which are often limited.
- 5) Risk: Because the projects are unique and involve uncertainty they always carry degree of risk.
- 6) Interrelated Activities : A project consists series of a tasks that are linked; the output of one task is often the input for the next.

Q2 write a short note on PMBOK

PMBOK stands for Project Management Body of Knowledge. It is globally recognized standard and a foundational guide for the project management institute, it provides the essential terminology guidelines for managing a Project successfully.

The PMBOK framework organizes project management into two main categories Process and Knowledge Areas.

1. Integration Management.
2. Scope Management
3. Schedule Management.
4. Cost Management
5. Quality Management
6. Resource Management
7. Communication Management
8. Risk Management
9. Process Management

Importance Of PMBOK

- Standardization : it provides a common language for Project Managers worldwide ensuring everyone uses the same terms
- Proven Best practices : It doesn't tell you what processes are generally recognized as good practices for most projects.
- flexibility : It is a framework, not a rigid methodology. Manager can relate the processes to fit the specific needs of their project.

Q3] Explain ITPM.

ITPM stands for Information Technology Project Management. It is a specialized branch of project management that focuses on the planning, organizing and managing of resources to bring about the successful completion of specific IT goals and objectives.

While general project management applies to any industry (like construction or manufacturing) ITPM is unique because it deals specifically with hardware, software, networking and data management.

- Key component of ITPM
1. Scope Management Defining exactly what the software or system will do. In IT this is challenging because scope creep.
 2. Software Development Life Cycle (SDLC) ITPM integrates standard Project management with coding, Testing (Deployment).
 3. Risk Management Managing IT-specific risk like data breaches, systems crashes or hardware incompatibility.
 4. Quality Assurance : Ensuring the final software is bug-free and meets the user requirements through sign off testing.

Challenges of ITPM

- High complexity - IT projects often involve complex integration between different software systems and hardware.
- Rapid change : Technology evolves so fast that a project tools or goals might become outdated before the project is finished.
- Intangibility unlike a building you cannot see a software being built physically, which makes tracking progress more difficult.

Q4] What is the role of RFP? Describe its content and purpose?

An RFP is a formal document used by an organization to invite vendors to submit bids for a project. It is most commonly used when the project requirements are complex and the organization is looking for the most innovative or cost-effective solution from external experts.

Purpose

- i] Structural selection: It provides a standardized framework to compare different vendors objectively.
- ii] Competitive Bidding: It encourages vendors to offer their best services and lowest prices to beat competitors.
- iii] Defining scope: It forces the organization to finalize their project requirements before the work begins.
- iv] Risk mitigation: By reviewing a vendor's past performance and financial organization avoids choosing an unreliable partner.

Key components.

1. Statement of Work: Defines the specifications, tasks, milestones and deliverables the vendor must complete.

2. Technical Requirements: Lists the technology stack, security standards.
3. Administrative Information: Includes the deadline for submission, the point of contact & legal terms.

Types of Request documents.

RFI (Request for Information) A preliminary document used to scout the market & see what solutions are. **RFP (Request for Proposal)**: Used when the problem is known but the organization wants the vendor to propose the best solution.

RFQ (Request for Quotation) used when the solution is already known and the organization is simply looking for the lowest price.

RFT (Request for Tender): A highly formal, legal invitation to supply goods/services. Common in government projects.

Q5] what is feasibility study? Explain its types, contents and purpose.

A feasibility study is an evaluation conducted during the project initiation phase to determine if a project is technically possible, economically justified and legally permissible.

Types

1. **Technical**: Evaluation of whether the current technology (hardware, software and network) and the technical skills of the development team are sufficient to build the system.
2. **Economic feasibility**: A cost benefit analysis that determines if the financial gains and long-term benefits of the software outweigh the development and maintenance costs.
3. **Legal feasibility**: An assessment of whether the project conflicts with legal requirements such as data protection laws, software licensing or copyright.
4. **Operational feasibility**: A measure of how well the proposed system will solve business problem and whether it will be accepted and utilized by the end-user.
5. **Scheduled feasibility**: An analysis of whether the project can be completed within the required timeframe window.

contents of a feasibility Report

- Executive Summary : A High-level brief of the project and the final recommendation
 - System Description : Definition of the proposed softwares features and the problem it aims to solve
 - Resource Requirement : List of human resource hardware and software tools needed.
 - Financial Analysis : Estimates of ROI break-even points , and total cost of ownerships.
 - Risk Assessment : List of human resources hardware and software tools needed.
 - Financial Analysis : Estimates of ROI break-even points , and total cost of ownerships.
 - Risk Assessment : A clear suggestions based on the findings.
- Purpose.
- Minimize Risk : Identifies Potential "deal-breakers" early to avoid wasted investment.
 - Strategic Alignment : Ensure the software project aligns with the long-term goal of the orgs.
 - Resource allocation : Helps management decide how to distribute limited manpower and capital effectively

Q6] Write a short note on WBS.

Work Breakdown Structure (WBS) is a deliverable - oriented hierarchical decomposition of the total scope of work to be carried out by the project team.

Key characteristics

- Hierarchy: It starts with the final product and at the top and branches down into a specific tasks.
- Mutually Exclusive: There should be no overlap in work between different elements to avoid duplication of effort.
- Outcome-oriented: It focuses on deliverables (eg "Database Design") rather than just a list of actions.

Structure of levels

- 1) Level 1: The entire project (eg "New E-commerce platform")
- 2) Level 2: Major Deliverables or Project phases (eg: frontend Backend Testing)
- 3) Level 3: The lowest level where work and schedules can be easily estimated.

Purpose and Benefit.

Improved estimation. It is much easier to estimate the cost and time for a small task than for an entire project.

Accountability: Assigns specific responsibilities to team members for clearly defined work packages.

Progress tracking: Provides a baseline for monitoring project status and identifies bottlenecks early.

Clarity: Prevents "Scope Creep" by defining exactly what needs to be built.

Q7] Consider a XYZ company under take a project to computerized working of ABC city bank then Develop WBS.

1. Project Management

Project Planning and scheduling
Stakeholder Meetings and communication
Risk management and Quality Assurance.

2. Functional Requirements Analysis

Study of Existing Manual Banking workflow
functional Requirements Gathering
Security and Compliance Requirements.

- 3 Infrastructure and hardware setup
Server room setup and networking
Procurement of Terminal.
Installations of UPS & Backup system.
- 4 Software Development.
Database Design
Core Banking Module
Customer Onboarding and KYC
Transaction Processing
Interest Calculation Engine
- 5 Data Migration
Extraction of Data from Manual ledgers
Data Cleaning and validation
Migration to New SQL.
- 6 Testing
Unit Testing of Modules
Security and Penetration Testing
User Acceptance Testing.
- 7 Deployment.
Staff training & module
System Go Live
Post implementation support and
Maintenance.